

TP1

Cheby

$$\omega_c = 3000\pi \quad \omega_s = 9000\pi$$

$$\Omega\omega = 3000\pi$$

$$\omega_s = 3$$

$$\alpha_{\max} = 0,5 \text{ dB} \quad \alpha_{\min} = 16 \text{ dB}$$

$$\omega_c = 1$$

$$\xi^2 = 10^{\frac{0,5}{10}} - 1 \approx 0,122$$

$$\alpha_{\min} = 10 \log(1 + \xi^2 \cdot \cosh^2(n \cdot \cosh^{-1}(\omega_s)))$$

$$n=1 \rightarrow 31,22 \text{ dB}$$

$$n=2 \rightarrow 15,59 \text{ dB}$$

$$n=3 \rightarrow 30,78 \text{ dB}$$

$$n=3$$

$$C_3(\omega) = 2\omega \cdot C_{n-1}(\omega) - C_{n-2}(\omega)$$

$$2\omega \cdot (2\omega^2 - 1) - \omega$$

$$C_3(\omega) = 4\omega^3 - 3\omega$$

entonces

$$\frac{1}{1 + \xi^2 \cdot C_3(\omega)^2} = \frac{1}{1 + \xi^2 \cdot (16\omega^6 - 24\omega^4 + 9\omega^2)} \approx \frac{\frac{1}{16\xi^2}}{\omega^6 - \frac{3}{2}\omega^4 + \frac{9}{16}\omega^2 + \frac{1}{16\xi^2}}$$

Pasando a dominio S

$$|T(s)|^2 = |T(j\omega)|^2 \Big|_{\omega=j} = \frac{\frac{1}{16\xi^2}}{\frac{s^6}{j^6} - \frac{3}{2} \cdot \frac{s^4}{j^4} + \frac{9}{16} \frac{s^2}{j^2} + \frac{1}{16\xi^2}}$$

$$\begin{aligned} j^6 &= -1 \\ j^4 &= 1 \\ j^2 &= -1 \end{aligned}$$

$$|T(s)|^2 = \frac{\frac{1}{16\xi^2}}{-s^6 - \frac{3}{2}s^4 - \frac{9}{16}s^2 + \frac{1}{16\xi^2}}$$

Esto se implementaría con un 2º orden y 1º orden

$$T(s) = k \cdot \frac{A}{s+A} \cdot \frac{C}{s^2+Bs+C}$$

Formando en conjunto un 3º orden

$$T(s) = D \cdot \frac{G}{s^3 + s^2E + sF + G}$$

entonces

$$T(s) \cdot T(-s) = \frac{D \cdot G}{-s^3 + s^2E - sF + G} \cdot \frac{D \cdot G}{s^3 + s^2E + sF + G}$$

$$T(s) \cdot T(-s) = \frac{D^2 \cdot G^2}{-s^6 - s^4 F \cdot 2 + s^4 E^2 + 2 \cdot s^2 E G - s^2 F^2 + G^2}$$

$$T(s) \cdot T(-s) = \frac{D^2 \cdot G^2}{s^6 \cdot (-1) + s^4 \cdot (E^2 - 2F) + s^2 (2 \cdot E \cdot G - F^2) + G^2}$$

Igualando a expresión de cheby

$$\frac{D^2 \cdot G^2}{s^6 \cdot (-1) + s^4 \cdot (E^2 - 2F) + s^2 (2 \cdot E \cdot G - F^2) + G^2} = \frac{\frac{1}{16\xi^2}}{s^6 \cdot (-1) + s^4 \cdot \left(-\frac{3}{2}\right) + s^2 \cdot \left(-\frac{9}{16}\right) + \frac{1}{16\xi^2}}$$

$$\begin{cases} G^2 = \frac{1}{16\xi^2}, D^2 = 1 \\ E^2 - 2F = -\frac{3}{2} \textcircled{I} \\ 2EG - F^2 = -\frac{9}{16} \textcircled{II} \end{cases}$$

$$G = \frac{1}{4\xi} \approx 0.7157$$

$$\textcircled{I} \left(E^2 + \frac{3}{2}\right) \frac{1}{2} = F \rightarrow F^2 = \left(\frac{E^2}{2} + \frac{3}{4}\right)^2$$

$$F^2 = \frac{E^4}{4} + \frac{2E^2}{2} \cdot \frac{3}{4} + \frac{9}{16}$$

$$\textcircled{II} \quad 2 \cdot E \cdot G - \frac{E^4}{4} - E^2 \cdot \frac{3}{4} - \frac{9}{16} = -\frac{9}{16}$$

$$E^4 \left(-\frac{1}{4}\right) + E^2 \left(-\frac{3}{4}\right) + 2 \cdot E \cdot \frac{1}{4\xi} = 0$$

$$E^3 \left(-\frac{1}{4}\right) + E \left(-\frac{3}{4}\right) + \frac{1}{2\xi} = 0 \quad (\text{con calc})$$

$$E_1 = 1.252912973$$

$$\begin{matrix} E_2 \\ E_3 \end{matrix} \quad \text{no conv.}$$

$$F = 1.534895459$$

con estos valores. factorizo para hallar los 2 filtros 2do y 1er orden
a utilizar

$$T(s) = K \cdot \frac{A}{s+A} \cdot \frac{C}{s^2+Bs+C} = \frac{K \cdot A \cdot C}{s^3 + Bs^2 + Cs + s^2A + BAS + CA}$$

$$T(s) = \frac{K \cdot A \cdot C}{s^3 + s^2(B+A) + s(AB+C) + AC} = \frac{G}{s^3 + s^2E + sF + G}$$

$$\begin{cases} A \cdot C = G, & \textcircled{I} \\ B+A = E & \textcircled{II} \\ AB+C = F & \textcircled{III} \end{cases}$$

$$\begin{aligned} \textcircled{I} \quad C &= \frac{G}{A} \\ \textcircled{II} \quad B &= E - A \\ \textcircled{III} \quad A \cdot (E - A) + \frac{G}{A} &= F \end{aligned}$$

$$-A^2 + EA + \frac{G}{A} = F$$

$$-A^3 + EA^2 - F \cdot A + G = 0$$

con G, F y E en calcul...

$$\underline{A_1 = 0.6264564863}$$

$$A_2 \text{ y } A_3 \notin$$

$$\underline{B = 0.6264564863} \quad B = A$$

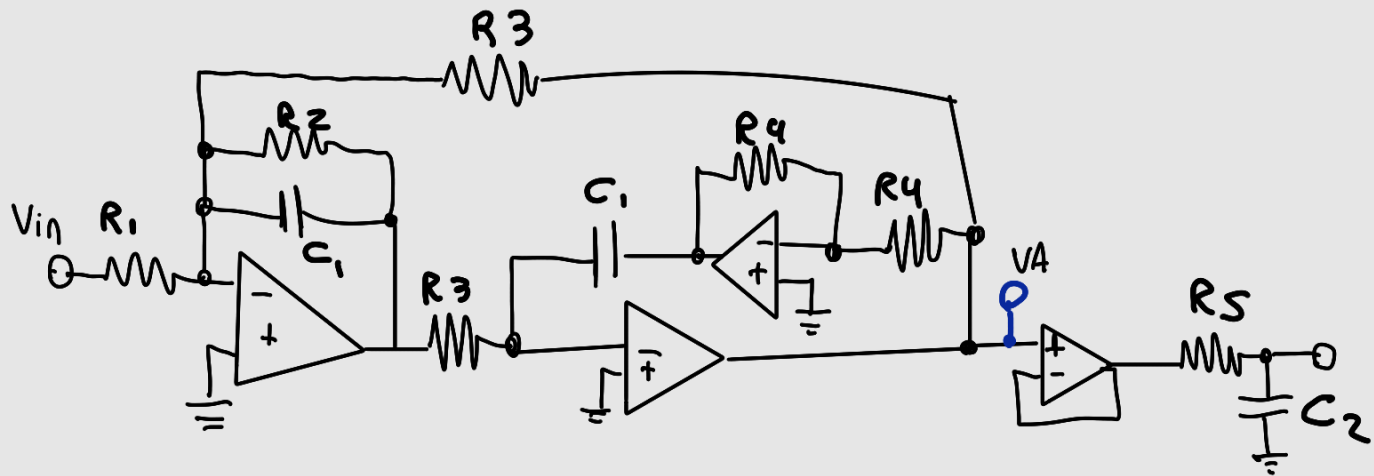
$$\underline{C = 1.142449729}$$

Finalmente, se llega a...

$$T(s) = \frac{0,6264565}{s + 0,6264565} \cdot \frac{1,14245}{s^2 + s \cdot 0,6264565 + 1,14245}$$

$$T(s) = \frac{0,71569}{s^3 + s^2 \cdot 1,252913 + s \cdot 1,5349 + 0,71569}$$

Un circuito posible podría ser



Donde $\frac{V_2}{V_{in}}$ se podría obtener como:

$$\frac{V_2(s)}{V_{in}(s)} = T_1(s) = -\frac{G_1}{G_3} \cdot \frac{\left(\frac{G_3}{C_1}\right)^2}{s^2 + s \cdot \frac{G_2}{C_1} + \left(\frac{G_3}{C_1}\right)^2}, \quad \begin{aligned} K_i &= -\frac{G_1}{G_3} \\ \omega_{0i} &= \frac{G_3}{C_1} \\ Q_i &= \frac{G_2}{G_3} \end{aligned}$$

Dejamos Todo en función de C_1

$$\omega_{0i} \cdot C_1' = G_3' \quad \text{I}, \quad G_2' = \frac{\omega_{0i} \cdot C_1'}{Q_i} \quad \text{II}, \quad -G_1' = K_i \cdot \omega_{0i} \cdot C_1' \quad \text{III}$$

NOTA: TOMAMOS

$K_1 = K_2 = -1$ para

que entre sí se anulen

Por el lado de $T_2(s)$..

$$\frac{V_o}{V_a} = T_2(s) = \frac{\frac{1}{R_5 C_2}}{s + \frac{1}{R_5 C_2}} = \frac{0,6264565}{s + 0,6264565}$$

donde

$$G_5' = \omega_{02} \cdot C_2' \quad \text{IV}$$

PARA DESNORMALIZAR, sé que

$$\omega = 3000\pi$$

Elijo C_1 y C_2 utilizables... $C_1 = C_2 = 10 \text{ nF}$
DESNORMALIZO ω_0

$$T_1(s) = \frac{1,14245}{s^2 + s \cdot 0,6264565 + 1,14245} \rightarrow T_1(s) = \frac{101,48 \cdot 10^6 \frac{1}{s}}{s^2 + s \cdot 5,904 \cdot 10^3 + 101,48 \cdot 10^6}$$

$$T_2(s) = \frac{5,904 \cdot 10^3 \frac{1}{s}}{s + 5,904 \cdot 10^3 \frac{1}{s}}$$

Retomando ecuaciones (I) - (IV).

$$G_3 = \sqrt{101,48 \cdot 10^6} \cdot 10 \text{ nF} = 1,007 \cdot 10^{-4} \rightarrow R_3 = 9926,8 \Omega$$

$$G_2 = 5,904 \cdot 10^3 \frac{1}{s} \cdot 10 \text{ nF} = 5,9042 \cdot 10^{-7} \rightarrow R_2 = 16,9 \text{ k}\Omega$$

$$G_1 = k \cdot G_3 = G_3 \rightarrow R_1 = 9926,8 \Omega$$

$$G_4 = G_1 \rightarrow R_4 = 9926,8 \Omega$$

$$G_5 = 5,946 \cdot 10 \text{ nF} \rightarrow R_5 = 16,9 \text{ k}\Omega$$